

Supplementary Materials for: SPECTRUM: A Data-Driven, Component-Level Projection Framework for 21st-Century Sea-Level Rise

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1 Materials and Methods

Unless otherwise noted, all reported intervals are 90% Bayesian credible intervals (equal-tailed), and where 1σ is reported (e.g., the rheology correction factor), we state explicitly that it is a standard deviation.

1.1 Observational datasets

All component models are calibrated against published observational products (Table S1). We use:

- **Global mean sea level:** NASA satellite altimetry (TOPEX/Jason, 1993–2025; *Beckley et al.*, 2017; *Hamlington et al.*, 2020), with the Frederikse et al. (2020) tide-gauge reconstruction (1900–2018) and the Dangendorf et al. (2024) reconstruction (1900–2021) for pre-satellite validation. None of these records enters any component calibration.
- **Ocean thermal expansion:** NOAA thermosteric sea level (doi:10.7289/V53F4MVP), 0–700 m (1955–2025) and 0–2000 m (2005–2025) (*Levitus et al.*, 2012). Both records are scaled by 1.14 before fitting. This corrects the 14% low bias that *Li et al.* (2022) find in thermosteric trends computed with linear vertical interpolation. A residual structural uncertainty of 5% of the corrected signal is retained. The 700–2000 m layer is carried by the 0–2000 m record from 2005 onward and by the deep layer of the two-layer model at earlier times. A constant $0.07 \pm 0.04 \text{ mm yr}^{-1}$ (*Purkey and Johnson*, 2010; *Desbruyères et al.*, 2016) is added over the full record for the contribution below 2000 m. Two reconstructions of global-mean steric sea level are withheld for validation, from Frederikse et al. (2020) (1900–2018) and Dangendorf et al. (2024) (1900–2021). Over 1957–2018 the Frederikse et al. reconstruction draws on the same archive of in situ profiles as the NOAA record (*Levitus et al.*, 2012; *Ishii et al.*, 2017; *Cheng and Zhu*, 2016), so it is not fully independent of it there. The Dangendorf et al. reconstruction uses no in situ ocean temperature or salinity. Its steric term is the residual of a sea-level budget whose barystatic input is from *Frederikse et al.* (2020). Neither reconstruction carries the 1.14 scaling.
- **Glaciers:** GlaMBIE global consensus (The GlaMBIE Team 2025), combining glaciological field measurements, satellite altimetry, DEM differencing, and GRACE/GRACE-FO gravimetry across all 19 RGI regions. We use 2000–2023 (24 annual points) for calibration.
- **Greenland:** *Mouginot et al.* (2019) SMB and discharge (1972–2018), of which the 44 years from 1975 enter the discharge fit, the earlier years being excluded because the lagged ocean forcing is undefined before then. *Mankoff et al.* (2021) discharge (1986–2023) and the IMBIE-3 Greenland record (1972–2023; *Otosaka et al.* (2026)) are withheld for validation. Subsurface ocean temperature from EN4 v4.2.2 with g10 bias corrections (200–500 m, Greenland peripheral waters, 1970–2021; *Good et al.* (2013)). Surface temperature from Berkeley Earth (*Rohde and Hausfather*, 2020).

- **Antarctica:** IMBIE-3 mass balance (1979–2023, 45 annual points per region; *Otosaka et al.* (2026)), partitioned into West Antarctica, East Antarctica, and the Antarctic Peninsula.
- **Terrestrial water storage:** *Frederikse et al.* (2020) reconstruction, used only from 2003 onward, the period over which it derives from GRACE and GRACE-FO observations rather than from models. Prior to 2003, the reconstructed TWS curves are largely unconstrained by independent observations.
- **Temperature scenarios:** IPCC AR6 assessed GMST projections (SSP1-2.6, SSP2-4.5, SSP3-7.0, SSP5-8.5; *Lee et al.* (2021)).

Table S1: Datasets used in the component framework and the role each plays. Records marked *validation* are withheld from every component calibration, including observed GMSL itself. A complete dataset inventory including versions, spatial domains, and processing details is provided in `data/dataset_inventory.xlsx`.

Component	Dataset	Span	Role	Reference
GMSL	NASA altimetry	1993–2025	Validation	Beckley et al. (2017)
GMSL	Frederikse reconstruction	1900–2018	Validation	Frederikse et al. (2020)
GMSL	Dangendorf reconstruction	1900–2021	Validation	Dangendorf et al. (2024)
Thermosteric	NOAA thermosteric, 0–700 m	1955–2025	Calibration	Levitus et al. (2012)
Thermosteric	NOAA thermosteric, 0–2000 m	2005–2025	Calibration	Levitus et al. (2012)
Thermosteric	Frederikse steric	1900–2018	Validation	Frederikse et al. (2020)
Thermosteric	Dangendorf sterodynamic	1900–2021	Validation	Dangendorf et al. (2024)
Glaciers	GlaMBIE global consensus	2000–2023	Calibration	The GlaMBIE Team (2025)
Greenland	Mouginot discharge & SMB	1972–2018	Calibration	Mouginot et al. (2019)
Greenland	EN4 subsurface ocean T	1970–2021	Forcing	Good et al. (2013)
Greenland	Mankoff PROMICE discharge	1986–2023	Validation	Mankoff et al. (2021)
Greenland	IMBIE-3 Greenland	1972–2023	Validation	Otosaka et al. (2026)
Peninsula	IMBIE-3 Peninsula	1979–2023	Calibration	Otosaka et al. (2026)
EAIS	IMBIE-3 East Antarctica	1979–2023	Diagnostic	Otosaka et al. (2026)
WAIS	IMBIE-3 West Antarctica	1979–2023	Calibration	Otosaka et al. (2026)
TWS	Frederikse TWS (GRACE era)	2003–2018	Forcing	Frederikse et al. (2020)
Temperature	Berkeley Earth surface T	1850–2024	Forcing	Rohde & Hausfather (2020)
Temperature	IPCC AR6 GMST projections	2015–2099	Forcing	Lee et al. (2021)
Projections	IPCC AR6 FACTS	2020–2150	Comparison	Garner et al. (2023)

1.2 Variables and parameters

Table S2 lists every symbol that appears in the main text, with a plain-language description, the value where one is assigned, units, and the source of any assigned value. Symbols used only in this supplement, including those of the value-of-information analysis, are defined where they appear there and are not repeated here.

Table S2: **Symbols used in the main text.** The value column gives the value of parameters that are assigned rather than inferred. Quantities we calibrate are marked *calibrated* and their posterior medians and 90% credible intervals are reported in the main text, since a fitted value is not a prescribed one. Model variables that carry no single value are marked with a dash. Where one symbol carries more than one meaning, each meaning is listed separately and its context named. SLE denotes sea-level equivalent.

Symbol	Description	Value	Units	Source
General				
t	Time	—	yr	—
T	Global mean surface temperature (GMST) anomaly, referenced to 1850–1900	—	°C	<i>Rohde and Hausfather (2020)</i>
ΔT	GMST anomaly forcing the surface-mass-balance models	—	°C	—
t_0	Study baseline year, to which cumulative contributions are referenced in the component models	2000	yr	—
t_0	Forecast origin, the most recent year of satellite altimetry in the blended forecast	2025	yr	—
i	Index over observations, outlet glaciers, or Monte Carlo samples	—	—	—
z	Depth below the sea surface	—	m	—
Greenland Ice Sheet discharge				
D	Greenland solid-ice discharge, as a contribution to sea level	—	mm SLE yr ^{−1}	—
H_D	Cumulative Greenland discharge contribution to GMSL	—	mm SLE	—
γ	Sensitivity of discharge to sub-surface ocean temperature	calibrated	mm SLE yr ^{−1} °C ^{−1}	this study
r_0	Background discharge rate at the baseline ocean temperature	calibrated	mm SLE yr ^{−1}	this study
δ	Lag between ocean warming and the discharge response	5, selected by BIC	yr	this study
T_{ocean}	Subsurface ocean temperature anomaly, 200–500 m, in Greenland’s peripheral waters	—	°C	<i>Good et al. (2013)</i>
Dimensional scaling for γ				
$\tilde{\gamma}$	Order-of-magnitude estimate of γ from the scaling argument	0.05–1.5	mm SLE yr ^{−1} °C ^{−1}	this study
$\Delta \dot{m}_i$	Submarine melt rate of outlet glacier i	—	m yr ^{−1}	—

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Symbol	Description	Value	Units	Source
κ_i	Melt sensitivity to ocean thermal forcing, averaged over the calving face	36–104, median 60	$\text{m yr}^{-1} \text{ } ^\circ\text{C}^{-1}$	<i>Xu et al.</i> (2013); <i>Rignot et al.</i> (2016); <i>Slater and Straneo</i> (2022); <i>Wood et al.</i> (2021)
μ_i	Ratio of the total ice-flux perturbation to the direct submarine melt flux	3–10	—	<i>Enderlin and Howat</i> (2013); <i>Wood et al.</i> (2021)
$A_{f,i}$	Submerged calving-front area of outlet glacier i	0.9–2.7	km^2	<i>Rignot et al.</i> (2016)
$\Delta\dot{Q}_i$	Ice-flux perturbation from outlet glacier i	—	Gt yr^{-1}	—
N	Number of marine-terminating outlet glaciers resolved in the record	219	—	<i>Mouginot et al.</i> (2019)
ρ_{ice}	Density of ice	900	kg m^{-3}	—
ρ_{fw}	Density of freshwater	1000	kg m^{-3}	—
A_{ocean}	Global ocean area, setting the convention 1 mm SLE = 362.5 Gt	3.625×10^{14}	m^2	—
Thermosteric two-layer model				
H_u	Steric height of the upper ocean layer	—	mm	—
H_{ud}	Steric height of the upper 2000 m	—	mm	—
H_{full}	Full-depth thermosteric contribution to GMSL	—	mm	—
α_{sw}	Thermal expansion coefficient of seawater	—	$^\circ\text{C}^{-1}$	—
α'_{sw}	Derivative of α_{sw} with respect to temperature	—	$^\circ\text{C}^{-2}$	—
$\Delta\theta$	Ocean temperature anomaly profile	—	$^\circ\text{C}$	—
φ	Dimensionless shape factor of the temperature anomaly profile	—	—	—
h_u	Thickness of the upper layer	700	m	—
h_d	Lower bound of the deep layer	2000	m	—
S_u	Thermal state of the upper layer, the amplitude of its temperature anomaly	—	$^\circ\text{C}$	—
S_d	Thermal state of the deep layer	—	$^\circ\text{C}$	—
a	Quadratic expansion coefficient of the upper layer, from the temperature dependence of α_{sw}	calibrated	$\text{mm } ^\circ\text{C}^{-2}$	this study
b_u	Linear expansion coefficient of the upper layer	calibrated	$\text{mm } ^\circ\text{C}^{-1}$	this study
b_d	Linear expansion coefficient of the deep layer	calibrated	$\text{mm } ^\circ\text{C}^{-1}$	this study

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Symbol	Description	Value	Units	Source
ϵ_u	Instrument drift absorbed during calibration	calibrated	mm yr ⁻¹	this study
τ_u	Relaxation time of the upper layer toward GMST	calibrated	yr	this study
τ_d	Relaxation time of the deep layer toward the upper layer, held fixed	150	yr	<i>Geoffroy et al.</i> (2013)
$\dot{H}_b$	Rate of steric expansion below 2000 m, adopted rather than calibrated	0.07 ± 0.04	mm yr ⁻¹	<i>Purkey and Johnson</i> (2010)
t_{ref}	Reference year for the below-2000 m term	2000	yr	—
σ_{extra}	Noise term absorbing misfit beyond the reported observational error	calibrated	mm	this study
ϕ	Vector of the six fitted thermosteric parameters	—	—	—
$\mathcal{J}$	Objective function, twice the negative log-posterior up to a constant	—	—	—
r_i	Residual between the demeaned model and observation i	—	mm	—
σ_i	Total standard deviation of observation i	—	mm	—
μ_{τ_u}	Mean of the log-normal prior on $\ln \tau_u$	$\ln 8$	$\ln(\text{yr})$	<i>Geoffroy et al.</i> (2013)
s_{τ_u}	Standard deviation of the log-normal prior on $\ln \tau_u$	0.5	—	<i>Geoffroy et al.</i> (2013)
Glaciers				
H_{gl}	Cumulative glacier contribution to GMSL	—	mm	—
b_{gl}	Sensitivity of the glacier loss rate to warming	calibrated	mm yr ⁻¹ °C ⁻¹	this study, prior from <i>Rounce et al.</i> (2023)
c_{gl}	Committed glacier loss rate, independent of further warming	calibrated	mm yr ⁻¹	this study
H_{gl_0}	Glacier level at the baseline year	calibrated	mm	this study
ε_{gl}	Residual of the glacier fit	—	mm	—
Σ_{gl}	Covariance matrix of the cumulative glacier record	—	mm ²	<i>The Glaciers and Ice Sheet Model Intercomparison Project Team et al.</i> (2025)
Antarctic Peninsula				
$\dot{H}_{\text{p}}$	Rate of the Antarctic Peninsula contribution	—	mm yr ⁻¹	—
a_{p}	Quadratic sensitivity to warming, set to zero because BIC favors the linear model	0	mm yr ⁻¹ °C ⁻²	this study

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Symbol	Description	Value	Units	Source
b_p	Sensitivity of the Peninsula rate to warming	calibrated	$\text{mm yr}^{-1} \text{ }^\circ\text{C}^{-1}$	this study
c_p	Committed Peninsula rate	calibrated	mm yr^{-1}	this study
H_{p0}	Peninsula level at the first year of the record (1979)	calibrated	mm	this study
$\sigma_{p,\text{extra}}$	Model-inadequacy term for annual scatter beyond the reported uncertainty	calibrated	mm	this study
East Antarctic Ice Sheet				
b_e	Sensitivity of the EAIS rate to warming	calibrated	$\text{mm yr}^{-1} \text{ }^\circ\text{C}^{-1}$	this study, prior from <i>Frieler et al.</i> (2015); <i>Ligtenberg et al.</i> (2013)
c_e	Committed EAIS rate	calibrated	mm yr^{-1}	this study
H_{e0}	EAIS level at the first year of the record	calibrated	mm	this study
West Antarctic Ice Sheet				
a	Acceleration of WAIS mass loss in the quadratic-in-time fit	calibrated	mm yr^{-2}	this study
$p(\text{S1})$	Probability assigned to WAIS Scenario 1, the control variable of the scenario mixture	0.90 slow, 0.10 fast	—	this study
λ	Shape parameter of the skew-normal S2 distribution in log space	3	—	this study, motivated by <i>Robel et al.</i> (2019)
β	Exponent of the power law in time followed by the S2 trajectory	2	—	this study
Surface mass balance				
$\dot{H}_{\text{smb}}$	Surface mass balance rate	—	Gt yr^{-1}	—
C_T	Linear sensitivity of Greenland SMB to GMST	$\mathcal{N}(-300, 80^2)$	$\text{Gt yr}^{-1} \text{ }^\circ\text{C}^{-1}$	<i>Fettweis et al.</i> (2020); <i>Noël et al.</i> (2021); <i>Glaude et al.</i> (2024)
C_{T^2}	Quadratic sensitivity of Greenland SMB, for ablation-zone expansion and the melt-albedo feedback	$\mathcal{N}(-50, 30^2)$	$\text{Gt yr}^{-1} \text{ }^\circ\text{C}^{-2}$	<i>Fettweis et al.</i> (2020); <i>Noël et al.</i> (2021); <i>Glaude et al.</i> (2024)
$\dot{M}_0$	Baseline Greenland SMB	380	Gt yr^{-1}	<i>Fettweis et al.</i> (2020); <i>Noël et al.</i> (2021)
C_T	Linear sensitivity of EAIS SMB to GMST, positive for mass gain by accumulation	60 ± 20	$\text{Gt yr}^{-1} \text{ }^\circ\text{C}^{-1}$	<i>Frieler et al.</i> (2015); <i>Ligtenberg et al.</i> (2013)
C_{T^2}	Quadratic sensitivity of EAIS SMB, set to zero	0	$\text{Gt yr}^{-1} \text{ }^\circ\text{C}^{-2}$	this study

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Symbol	Description	Value	Units	Source
$\dot{M}_0$	Baseline EAIS SMB, set to zero because IMBIE-3 shows EAIS near balance	0	Gt yr ⁻¹	<i>Otosaka et al.</i> (2026)
Greenland temperature chain				
$T_{\text{surface}}^{\text{Gr}}$	Greenland regional surface temperature anomaly	—	°C	<i>Rohde and Hausfather</i> (2020)
ξ	Slope of the surface-to-ocean transfer function, fitted by ordinary least squares	fitted, 0.148 ± 0.027	—	this study
χ	Intercept of the surface-to-ocean transfer function	fitted, -0.051 ± 0.031	°C	this study
ε_{go}	Zero-mean residual of the transfer function	—	°C	—
AA	Arctic amplification factor, the ratio of Greenland to global warming	2.5 ± 0.5	—	this study, from the observed ratio over 1970–2021
Forecast construction				
$\dot{H}_{\text{forecast}}$	Blended forecast rate of GMSL rise	—	mm yr ⁻¹	—
H_{forecast}	Forecast GMSL level	—	mm	—
$\dot{H}_{\text{obs}}$	Rate from the satellite-era quadratic fit to altimetry	—	mm yr ⁻¹	<i>Hamlington et al.</i> (2024); <i>Nerem et al.</i> (2022)
$\dot{H}_{\text{comp}}$	Rate of the summed calibrated components	—	mm yr ⁻¹	—
H_k	Level-space projection ensemble for component k	—	mm	—
w	Weight blending the observed and component rates	—	—	—
Θ	Standard logistic function	—	—	—
t_r	Midpoint year of the blend transition	2035 forecast, 2015 hindcast	yr	this study
τ_r	Width of the blend transition	5	yr	this study
Diagnostics				
$\dot{S}$	Rate of GMSL rise	—	mm yr ⁻¹	—
α	Transient sensitivity of the rate of GMSL rise to warming, $d\dot{S}/dT$	—	mm yr ⁻¹ °C ⁻¹	—
R^2	Coefficient of determination	—	—	—
χ^2/dof	Reduced chi-square, the error-weighted sum of squared residuals per degree of freedom	—	—	—

1.3 Component model posteriors

This section collects the parameter posteriors of every calibrated component model. Greenland discharge is fitted by weighted least squares over a discrete set of delays, so its posterior is a mixture of normals weighted by the Bayesian Information Criterion and is shown as

marginal densities. The remaining components are sampled by Markov chain Monte Carlo and are shown as corner plots, with marginal histograms on the diagonal and pairwise joint densities off it, so that the parameter trade-offs the marginals hide can be read directly. Posterior medians and their 90% credible intervals are reported in the main text, and they are the values carried into the projections.

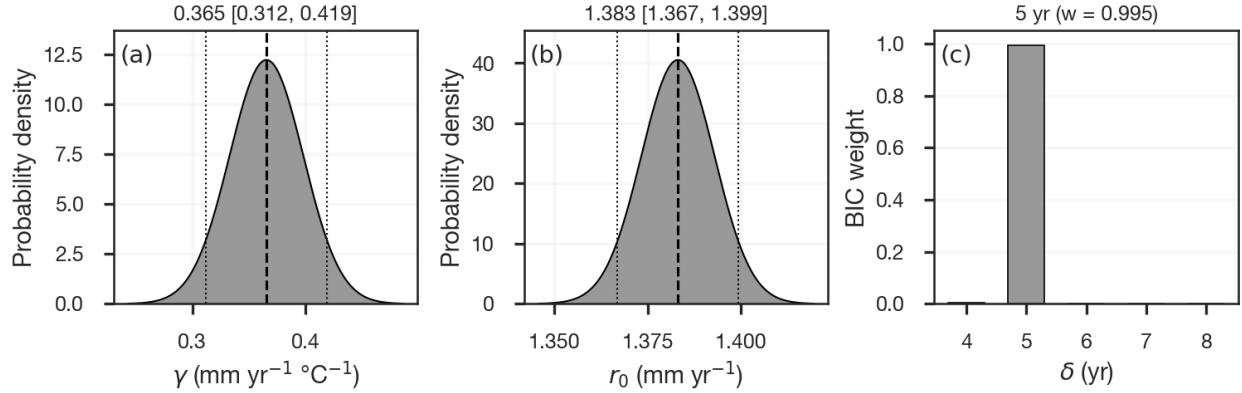


Figure S1: **Greenland discharge delay model parameter posteriors.** Marginal posteriors for the three parameters of the discharge delay model calibrated against *Mouginot et al.* (2019). **(a)** The sensitivity of discharge to cumulative delayed ocean thermal forcing, γ , and **(b)** the background discharge rate, r_0 . **(c)** The delay δ is resolved only to integer years, so its posterior is the set of weights over the candidate delays, which place 99.5% of the weight on $\delta = 5$ yr. Panel titles give the median with the 90% credible interval.

1.4 Variance decomposition

A variance decomposition using the law of total variance across the three policy-relevant SSPs (1-2.6, 2-4.5, 3-7.0) attributes 92.5% of total forecast variance to within-scenario uncertainty—predominantly WAIS (88.7% of total variance)—and 7.5% to between-scenario differences. Cross-component covariances are small (+0.0064 m², representing 2.0% of total variance). Most of this covariance comes from the warming path that all components share within each ensemble member, not from the calibrations, which are independent.

1.5 Temperature scenarios

The component projections are driven by the IPCC AR6 projections of global mean surface temperature for each SSP (*Lee et al.*, 2021). Each SSP has a median trajectory and a 5–95% range, which comes mostly from uncertainty in the response of the climate to a given emissions pathway. A world that warms faster than the median tends to stay warmer for the whole century. We therefore treat this range as a set of warming paths rather than as independent noise in each year.

Each ensemble member k is assigned one value z_k drawn from a standard normal distribution. Its temperature is the median trajectory plus an offset $\delta_k(t) = z_k [h(t) - h(t_0)]$ for $t > t_0$, and zero before t_0 . When $z_k > 0$, $h(t)$ is the distance from the median to the 95th

percentile divided by 1.645. When $z_k < 0$, it is the distance from the 5th percentile to the median divided by 1.645. This keeps the asymmetry of the AR6 range. We set $t_0 = 2024$ because warming up to the present is observed, so the offset adds spread only to future warming. As a result, the spread at 2100 is somewhat narrower than the full AR6 range. For SSP2-4.5 the upper half-width at 2099 is 0.64°C rather than 0.82°C .

Member k uses the same z_k in every component and every SSP. Each member therefore represents one world, in which the ocean, the glaciers, and the ice sheets all respond to the same warming. The z_k are drawn with their own random seed, so the parameter draws of each component do not change. Sampling the warming range widens the 2100 intervals of the temperature-driven components, most strongly for thermal expansion. At 3°C its 90% interval grows from 87 to 156 mm.

1.6 Component-level skill metrics

Table S3 collects the goodness-of-fit statistics for every component fitted to data. We report reduced chi-square rather than a coefficient of determination. Every component is calibrated against a cumulative record whose variance is dominated by its trend, so R^2 in that space approaches unity for any model that reproduces the trend at all and says little about the quality of the fit. A straight line in time, with no temperature dependence whatsoever, scores 0.983 against the 0–2000 m thermosteric record.

Reduced chi-square compares the residuals with the uncertainty the calibration data are reported to carry, so a value near unity indicates both that the model reproduces the observations to within their stated errors and that the error model used in the likelihood is well specified. The normalization is not identical in every case, and we note where it differs. Glaciers, the Antarctic Peninsula, and EAIS are fitted to cumulative records whose errors are serially correlated, and their χ^2 is therefore evaluated against the full covariance of the cumulative record built from the reported annual uncertainties, not against a diagonal approximation, which would understate the residual scatter that the reported errors can absorb. All three sit between 0.91 and 1.06. The Peninsula reaches that value only with a fitted model-inadequacy term $\sigma_{p,\text{extra}}$, described in the Antarctic Peninsula section of the main text. Without $\sigma_{p,\text{extra}}$, the reduced chi-square is 2.64, so the IMBIE-3 Peninsula data carries real annual scatter beyond its reported per-year uncertainty. Glaciers and EAIS do not need model-inadequacy terms. For the thermosteric model the quoted value includes the fitted excess-noise term σ_{extra} in the point variance and is normalized by the number of data points rather than by the degrees of freedom, so it sits slightly below the corresponding χ^2/dof . The 0–2000 m value of 0.36 indicates that the 21-year record is fitted to well inside its reported error. The Greenland discharge value of 1.83 is the generalized chi-square of the demeaned cumulative fit before the error scale is recalibrated. It is the quantity by which the parameter covariance is subsequently inflated, so it is a statement about the reported errors being too small rather than about the model missing the data. It has 41 degrees of freedom (= 44 fitted points minus one for the demeaning of the cumulative record and two for the fitted parameters). The *Mouginot et al.* (2019) discharge record runs from 1972, but the delayed ocean forcing is undefined before 1975 because EN4 begins in 1970 and the selected lag is 5 yr, so the first three years of the discharge record do not enter the fit.

Two components need further comment. WAIS is fitted in time rather than in tem-

perature, so no statistic comparable with the other rows is available for it. A weighted least-squares cross-check on the same record, using diagonal rather than correlated weights, gives $\chi^2 = 38.8$ on 44 annual points with three free parameters, and favors a cubic in time over the quadratic by $\Delta\text{BIC} = -4.1$. We retain the quadratic because the purpose of the WAIS fit is to set the starting rate and acceleration for the Scenario 1 extrapolation rather than to identify the polynomial order best supported by a 45-year record, and a cubic extrapolated to 2100 is governed by a term the observations constrain only weakly. EAIS is listed as a diagnostic because the East Antarctic Ice Sheet is close to balance over the IMBIE-3 record, leaving little signal for a temperature-dependent model to recover. The fit is consistent with the reported errors, but the projection adopts a literature surface-mass-balance sensitivity, $C_T = 60 \pm 20 \text{ Gt yr}^{-1} \text{ }^\circ\text{C}^{-1}$ (Frierler *et al.*, 2015; Ligtenberg *et al.*, 2013), and uses the regression only as a check on that value.

One further fit enters the framework without being a component model in its own right. The linear surface-to-ocean transfer function given in the main text, which carries SSP temperature trajectories into the Greenland discharge model, explains a modest fraction of the interannual variance, $R^2 = 0.38$ over 52 common annual values, with a residual standard deviation of $0.20 \text{ }^\circ\text{C}$ that is propagated into the projections. It is the weakest statistical link in the Greenland projection chain, and it is retained rather than replaced because the quantity it must supply is the multi-decadal trend in subsurface temperature rather than its year-to-year variability. Greenland SMB, by contrast, is not fitted. It is passed through as RACMO observations over the historical period and projected with literature sensitivities drawn from the spread between regional climate models (Fettweis *et al.*, 2020; Noël *et al.*, 2021; Glaude *et al.*, 2024), so no skill metric attaches to it.

These are all within-sample statistics, and on their own they establish only that each model is consistent with the record it was fitted to. The tests that bear on forecasting skill are out of sample. The thermosteric model is compared against the Frederikse *et al.* (2020) steric reconstruction, withheld from calibration, with an RMS difference of 2.6 mm over 1957–2018 and 3.8 mm before 1957, where that reconstruction is itself constrained by models rather than by subsurface observations. Greenland discharge is compared against the Mankoff *et al.* (2021) record (1986–2023) and the IMBIE-3 time series (Otosaka *et al.*, 2026), neither of which enters the fit. The model falls within error of both for discharge and for total mass balance. As noted in the main text, the Mankoff *et al.* (2021) comparison is the weaker of the two, because that record shares its underlying raw data with the Mouginot *et al.* (2019) calibration and differs from it in processing rather than in provenance. IMBIE-3 is the more informative test, being a combination of independent mass-balance estimates from satellite observations of ice flow, surface elevation, and gravimetry. The WAIS quadratic reaches $8.8 \pm 0.6 \text{ mm}$ at 2024 against the IMBIE-3 value of $9.0 \pm 0.6 \text{ mm}$. The tests that exercise the framework as a whole rather than one component at a time are budget closure, in which the summed components are compared with satellite altimetry measurements that no component calibration saw, and the hindcast of the level, rate, and transient sensitivity of GMSL since 1950. Both are presented in the Results of the main text.

Table S3: Goodness-of-fit for the component models fitted to data. For glaciers, the Peninsula, and EAIS, χ^2/dof is evaluated against the full covariance of the cumulative record. For the thermosteric model it is normalized by the number of data points rather than by the degrees of freedom and includes the fitted σ_{extra} in the point variance. For Greenland discharge it is the generalized chi-square before error-scale recalibration. Dataset references are given in Table S1. Greenland SMB and terrestrial water storage are not calibrated against data and are omitted.

Component	Calibration data	Reduced χ^2	Model
Thermosteric (0–700 m)	NOAA, 1955–2025 (71)	1.12	Two-layer, 6 fitted
Thermosteric (0–2000 m)	NOAA, 2005–2025 (21)	0.36	Joint with 0–700 m
Glaciers	GlaMBIE, 2000–2023 (24)	1.06	Linear rate– T
Greenland discharge	Mouginot, 1975–2018 (44)	1.83	Delay, 2 fitted
Antarctic Peninsula	IMBIE-3, 1979–2023 (45)	0.97	Linear rate– T
EAIS (diagnostic)	IMBIE-3, 1979–2023 (45)	0.91	Linear rate– T
WAIS	IMBIE-3, 1979–2023 (45)	—	Quadratic in time

2 Exceedance Probability by Threshold and Scenario

Table S4 reports the probability of exceeding selected global mean sea-level thresholds (relative to pre-industrial) by 2100 under three warming trajectories, for the slow-WAIS case, $p(\text{S1}) = 0.90$, and the fast-WAIS case, $p(\text{S1}) = 0.10$. Together these columns isolate the contribution of WAIS tail risk: the difference between them is the effect of shifting the scenario weight from very unlikely to very likely continued mass loss along the observed trajectory. The slow-WAIS column retains 10% weight on the accelerating branch, so probabilities there fall but do not vanish.

The tabulated values are the fractions of posterior samples above each threshold. Fig. 9b of the main text plots the same quantity as a continuous curve over sea level. Those curves are smoothed for display only, by replacing each sample with a Gaussian kernel whose width is one quarter of Scott’s rule for that ensemble, about 30 mm for the fast-WAIS case and 8 mm for the slow-WAIS case. At every threshold and warming trajectory in Table S4, the smoothed curves differ from the tabulated values by at most 0.5 percentage points.

3 Value of Information Analysis

The value of information (VoI) analysis quantifies the economic benefit of resolving the binary uncertainty about whether WAIS retreat remains slow (Scenario 1) or allows for the possibility of rapid retreat (Scenario 2). We compute the expected value of perfect information (EVPI), the perfect-information limit of the sample-information framework of *Raiffa and Schlaifer* (1961). No signal model is specified, so any test that discriminates the two scenarios imperfectly would return less than the values reported here.

Table S4: **Probability (%) of exceeding global mean sea-level thresholds by 2100.** Thresholds are relative to the pre-industrial baseline ($\approx 1850\text{--}1900$). The $p(S1) = 90\%$ column is the slow-WAIS case, which keeps 10% weight on the accelerating branch; the $p(S1) = 10\%$ column is the fast-WAIS case. Values are computed from posterior samples under each warming trajectory.

SLR threshold (m, pre-ind.)	Warming trajectory	SPECTRUM p(s1)=90%	SPECTRUM p(s1)=10%
0.5	2 °C	99.6	100.0
	3 °C	100.0	100.0
	4 °C	100.0	100.0
0.75	2 °C	24.0	70.7
	3 °C	88.7	97.7
	4 °C	99.5	100.0
1.0	2 °C	3.0	27.0
	3 °C	12.5	52.3
	4 °C	68.0	88.9
1.5	2 °C	1.0	8.6
	3 °C	1.4	12.2
	4 °C	2.9	21.1
2.0	2 °C	0.4	3.3
	3 °C	0.5	4.5
	4 °C	0.8	7.0

3.1 Decision framework

We consider an illustrative coastal adaptation decision where a planner must choose a defense height d (meters of sea-level protection) to minimize total expected costs, evaluated against the sea-level rise distribution S at a representative target year of 2100. For a given sea-level rise distribution S at 2100, the total annual cost of defense height d is

$$C(d) = C_{\text{adapt}}(d) + \mathbb{E}[D(\max(S - d, 0))], \quad (1)$$

where $C_{\text{adapt}}(d) = c_a \cdot d$ is the annualized adaptation cost and $\mathbb{E}[D(\cdot)]$ is the expected value of the residual damage D from SLR exceeding the defense height (overtopping).

3.1.1 Adaptation cost

The total capital cost of coastal adaptation is estimated at $C_{\text{cap}} = \$4 \text{ T/m}$ of GMSLR. This is derived from Tiggeloven et al. (2020), who estimate that 3.4% of the global coastline ($\sim 54,400 \text{ km}$ of ~ 1.6 million km total) warrants dike heightening under cost-benefit optimality, at a unit cost of $\$7 \text{ M}$ per km per meter of dike height (the median cost raising dikes in New Orleans post-Hurricane Katrina). This yields $\sim \$400 \text{ B/m}$ for dike raising alone along the optimal 3.4% of coastline. We apply a conservative $10\times$ multiplier to account for the full societal adaptation burden—including seawalls, nature-based solutions, managed retreat, and infrastructure relocation—across a broader fraction of the global coastline.

Because the cost function $C(d)$ (Eq. 1) combines adaptation costs with annual flood damages from Jevrejeva et al. (2018), both terms need to be expressed in consistent units. Adaptation is dominated by capital investment deployed over the planning horizon, so we conservatively annualize C_{cap} using the capital recovery factor (CRF):

$$c_a = C_{\text{cap}} \times \text{CRF}, \quad \text{CRF} = \frac{r(1+r)^n}{(1+r)^n - 1}, \quad (2)$$

where $r = 0.03$ is the real discount rate and $n = 75$ yr is the planning horizon. This gives $\text{CRF} \approx 0.034$ and an annualized adaptation cost of $c_a \approx \$135 \text{ B}/(\text{m} \cdot \text{yr})$ of GMSLR. The resulting $C(d)$ is in $\$/\text{yr}$, the EVPI is in $\$/\text{yr}$, and the present-value conversion (Eq. 6) produces a present-value total in $\$$.

3.1.2 Damage function

Residual flood damages are computed using the piecewise-linear global damage function of Jevrejeva et al. (2018), which maps SLR to annual global flood costs based on estimated asset exposure. For SLR beyond the tabulated range (1.80 m), we extrapolate linearly using the slope of the final segment ($\sim \$13.8 \text{ T}/\text{yr}$ per meter).

3.2 Two-world structure

The forecast distribution is decomposed into two conditional distributions, conditioned directly on A4 scenario membership rather than on a percentile-threshold proxy:

- **The S1 world** (S_{S1}) samples from the blended forecast conditional on WAIS Scenario 1 (`S1_status_quo`) representing future GMSLR.
- **The S2 world** (S_{S2}) samples from the full forecast conditional on WAIS Scenario 2 (`S2_fast_wais`), representing outcomes that allow for rapid WAIS retreat via MISI, with or without MICI.

Because the A4 framework comprises exactly these two scenarios, the two worlds partition it completely, but they are built differently. Every Monte Carlo sample’s WAIS scenario assignment is recorded at the point of sampling and carried through the component-summation and rate-space-blending pipeline as a per-sample label, and S_{S2} is the 1,805 samples of the blended forecast carrying the S2 label, an exact conditional subset rather than a percentile-threshold proxy. S_{S1} is not the 195 samples carrying the S1 label, which would be too few to resolve the tail; it is a separate 2,000-member ensemble in which WAIS is replaced by its S1 trajectory for every sample, with the non-WAIS components unchanged. The two branches therefore have different sample sizes, 2,000 and 1,805.

The VoI takes these two conditional worlds as its inputs and recombines them below with the prior p_s as the weight, so the slow- and fast-WAIS cases of the main text, at $p(\text{S1}) = 0.90$ and 0.10, are mixtures of these worlds rather than inputs to the calculation. The current prior probability of the S1 world is $p_s = 0.10$, the scenario weight of the fast-WAIS case presented in the main text.

3.3 EVPI calculation

For a given prior $p_s \equiv p(\text{S1})$ on the S1 world, the EVPI is the annual benefit of learning which scenario holds before choosing the defense height for coastal development.

3.3.1 Hedging cost (no information)

Without additional information, the planner hedges by choosing the single defense height d_{hedge}^* that minimizes the prior-weighted expected cost:

$$C_{\text{hedge}}^* = \min_d [p_s \cdot C_{\text{S1}}(d) + (1 - p_s) \cdot C_{\text{S2}}(d)], \quad (3)$$

where $C_{\text{S1}}(d) = C_{\text{adapt}}(d) + \mathbb{E}[D(\max(S_{\text{S1}} - d, 0))]$ and $C_{\text{S2}}(d)$ is defined analogously using S_{S2} .

3.3.2 Perfect information cost

With perfect information, the planner would tailor the defense height to whichever world is realized:

$$C_{\text{perfect}}^* = p_s \cdot \min_d C_{\text{S1}}(d) + (1 - p_s) \cdot \min_d C_{\text{S2}}(d). \quad (4)$$

3.3.3 Annual EVPI

The annual value of perfect information is the difference between the hedging cost and the perfect-information cost:

$$\text{EVPI}_{\text{ann}} = C_{\text{hedge}}^* - C_{\text{perfect}}^*. \quad (5)$$

This is the annual cost penalty of having to compromise on a single design rather than tailoring to the true state. It is non-negative by construction, because C_{hedge}^* minimizes a weighted sum of the two cost curves while C_{perfect}^* minimizes each of them separately. As a minimum over d of functions affine in p_s , C_{hedge}^* is concave in p_s , and subtracting the affine C_{perfect}^* leaves a concave function that vanishes at $p_s = 0$ and $p_s = 1$, so the maximum is interior. Its location is set by the geometry of the two cost curves and need not be at $p_s = 0.5$. Here the two worlds are strongly asymmetric, with optimal designs of 0.90 m under S1 against 2.95 m or more under S2, and the maximum falls at $p_s = 0.69$.

3.4 Discounting and the VoI surface

The annual EVPI is converted to a present value at the year of arrival by discounting the stream of benefits from the year information arrives (t_a) to the target year ($t_T = 2100$) at a real discount rate $r = 3\%$:

$$\text{VoI}(t_a, p_s) = \text{EVPI}_{\text{ann}}(p_s) \max\left(0, \frac{1 - (1 + r)^{-(t_T - t_a)}}{r}\right). \quad (6)$$

The annuity factor $[1 - (1 + r)^{-n}]/r$ captures the number of discounted years over which the improved decision yields benefits.

The VoI surface (Fig. 10c of the main text) is computed over a grid of arrival years ($t_a \in [2026, 2150]$) and prior probabilities ($p_s \in [0.01, 0.99]$). All calculations use the 3°C (SSP2-4.5) warming trajectory. The choice of warming trajectory has qualitatively little impact on the results given all the uncertainties involved. Note that our cost estimates are likely to be underestimates (see §3.1.1), but the takeaway message remains even if the true costs or damages are an order of magnitude lower. The main text quotes the maximum of this surface over both axes as of order US\$5 T. The exact maximum is US\$4.7 T, attained at $p_s = 0.69$ for information arriving in 2026. It is not the value at any single prior. At the $p(S1) = 0.90$ case adopted in the main text the surface gives US\$3.6 T, and at $p_s = 0.10$, the prior marked on the panel, it gives US\$0.8 T. The defense heights are searched on a grid capped at 2.95 m, which binds for the S2 world, so the values are biased low by roughly 6%; widening the grid leaves the location of the maximum unchanged.

3.5 Key assumptions and limitations

The VoI calculation is intended to be simple analysis to make a simple point, not to stand for detailed financial decisions. As such, we make several simplifying assumptions:

1. **Binary resolution.** Information perfectly resolves the slow/fast dichotomy. Partial information (e.g., updating p_s from 0.10 to 0.30 without full resolution) would yield lower VoI.
2. **Single decision.** The planner makes one irreversible design choice. In practice, adaptive management with staged decisions would reduce the hedging penalty, lowering VoI.
3. **Global aggregation.** Costs are aggregated globally. Local VoI would differ based on coastal exposure and elevation.
4. **Linear adaptation cost.** We assume C_{adapt} scales linearly with defense height. Economies or diseconomies of scale would modify the optimal designs.
5. **Discount rate.** The 3% real discount rate is standard for infrastructure but lower rates (reflecting intergenerational equity) would increase VoI. Higher discount rates, that would lower VoI, are unlikely given the long timeline of risk exposure from sea level rise.

Despite these simplifications, the analysis provides a rough order-of-magnitude estimate of the economic value of resolving WAIS uncertainty, supporting the argument that targeted investment in ice-sheet observations and process understanding has large expected returns.

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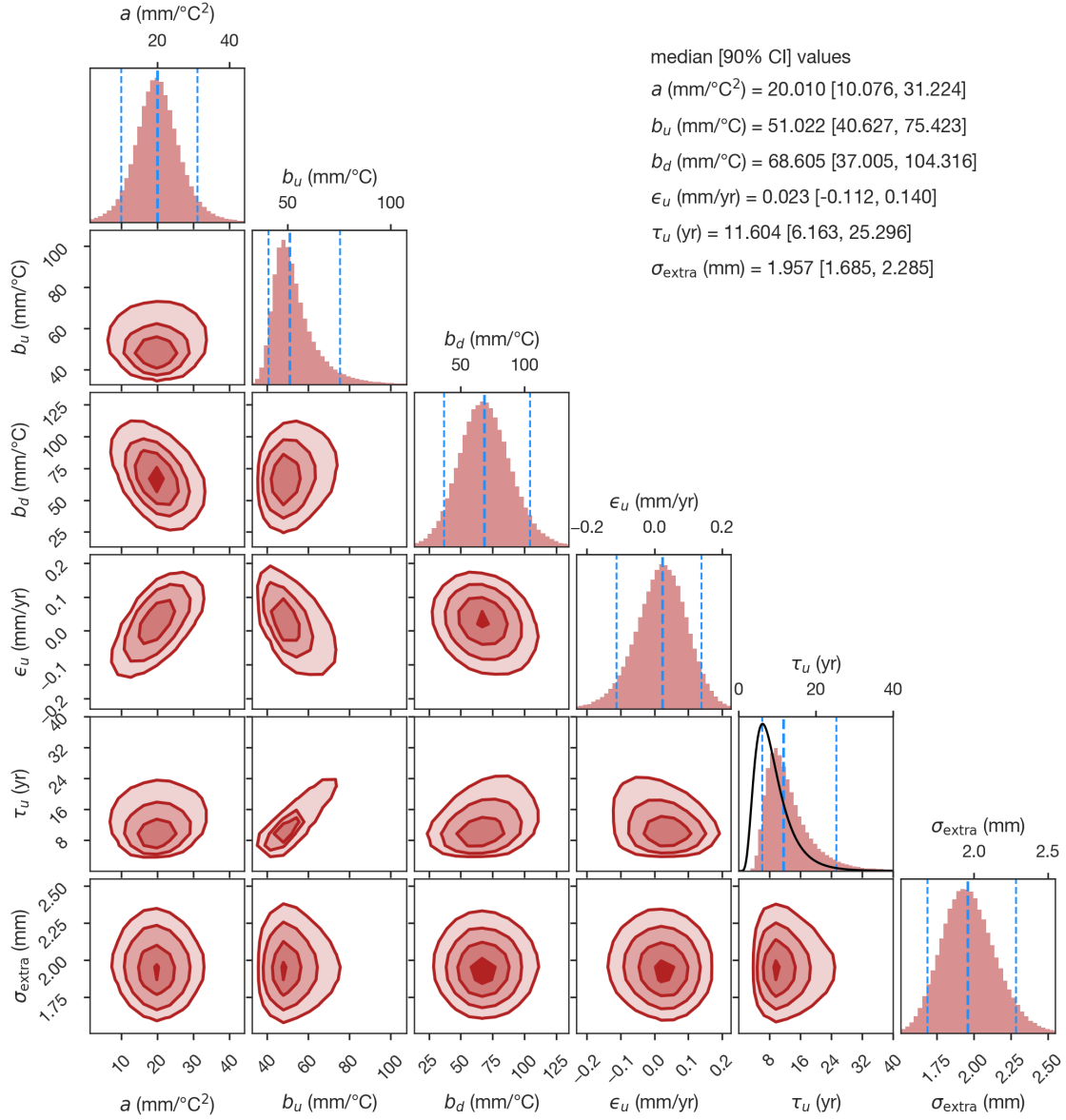


Figure S2: **Two-layer thermosteric model posterior correlation structure.** The joint posterior for the six parameters of the two-layer energy balance model fitted to NOAA 0–700 m and 0–2000 m thermosteric sea level data. Diagonal panels show marginal posterior histograms (red), with dashed vertical lines marking the median and 90% credible interval. Off-diagonal panels show the joint posterior density, with contours denoting 1, 2, and 3 standard deviations. Only τ_u has an informative prior (black curve) while the other five parameters have flat, uninformative priors (not shown).

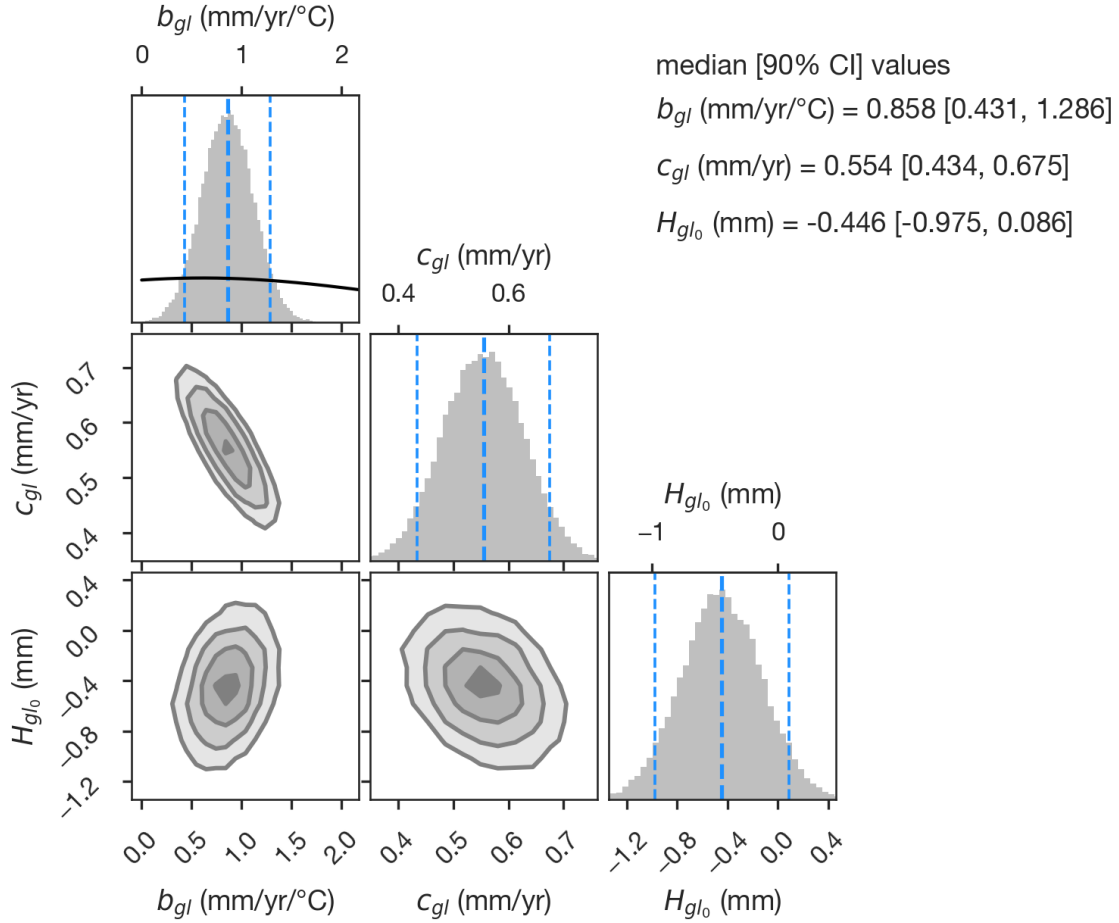


Figure S3: **Glacier model posterior correlation structure.** The joint posterior for the three parameters of the glacier model fitted to the GlaMBIE record (*The GlaMBIE Team et al.*, 2025), with the full covariance of the cumulative record in the likelihood. Diagonal panels show marginal posterior histograms (gray), with dashed vertical lines marking the median and 90% credible interval. Off-diagonal panels show the joint posterior density, with contours denoting 1, 2, and 3 standard deviations. Only b_{gl} has an informative prior (black curve), centered on the sensitivity of *Rounce et al.* (2023) and wide enough that the data set the posterior, while c_{gl} and H_{gl_0} have wide priors that are not shown.

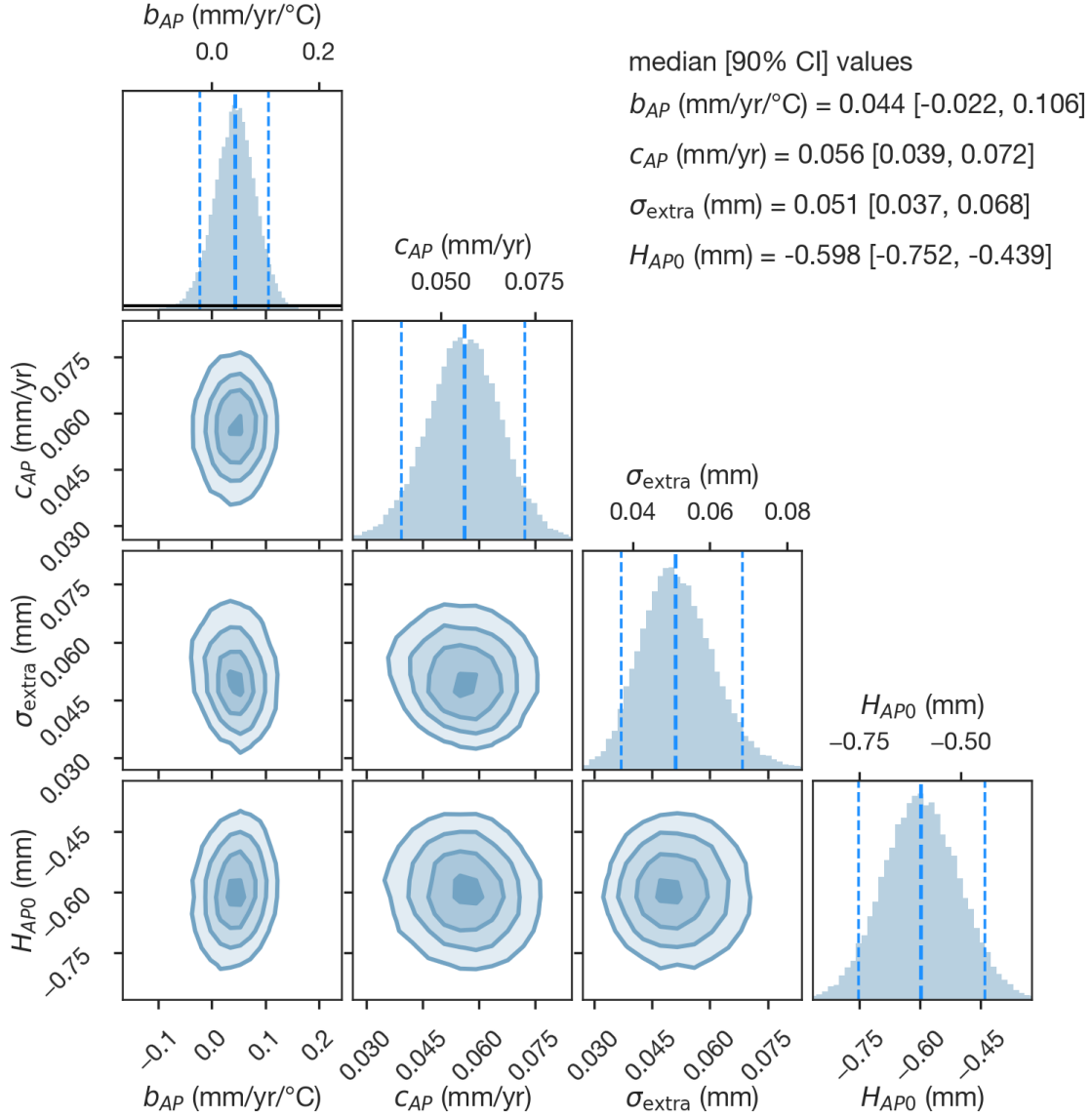


Figure S4: **Antarctic Peninsula model posterior correlation structure.** The joint posterior for the four parameters of the Peninsula model fitted to the IMBIE-3 Peninsula record (*Otosaka et al.*, 2026), with the full covariance of the cumulative record in the likelihood. The fourth parameter is the model-inadequacy term σ_{extra} , fitted here but not for the other components. Diagonal panels show marginal posterior histograms (blue), with dashed vertical lines marking the median and 90% credible interval. Off-diagonal panels show the joint posterior density, with contours denoting 1, 2, and 3 standard deviations. Only b_{AP} has a prior shown (black curve), a symmetric normal that is flat over the range the data support, while the remaining three parameters have wide priors that are not shown.

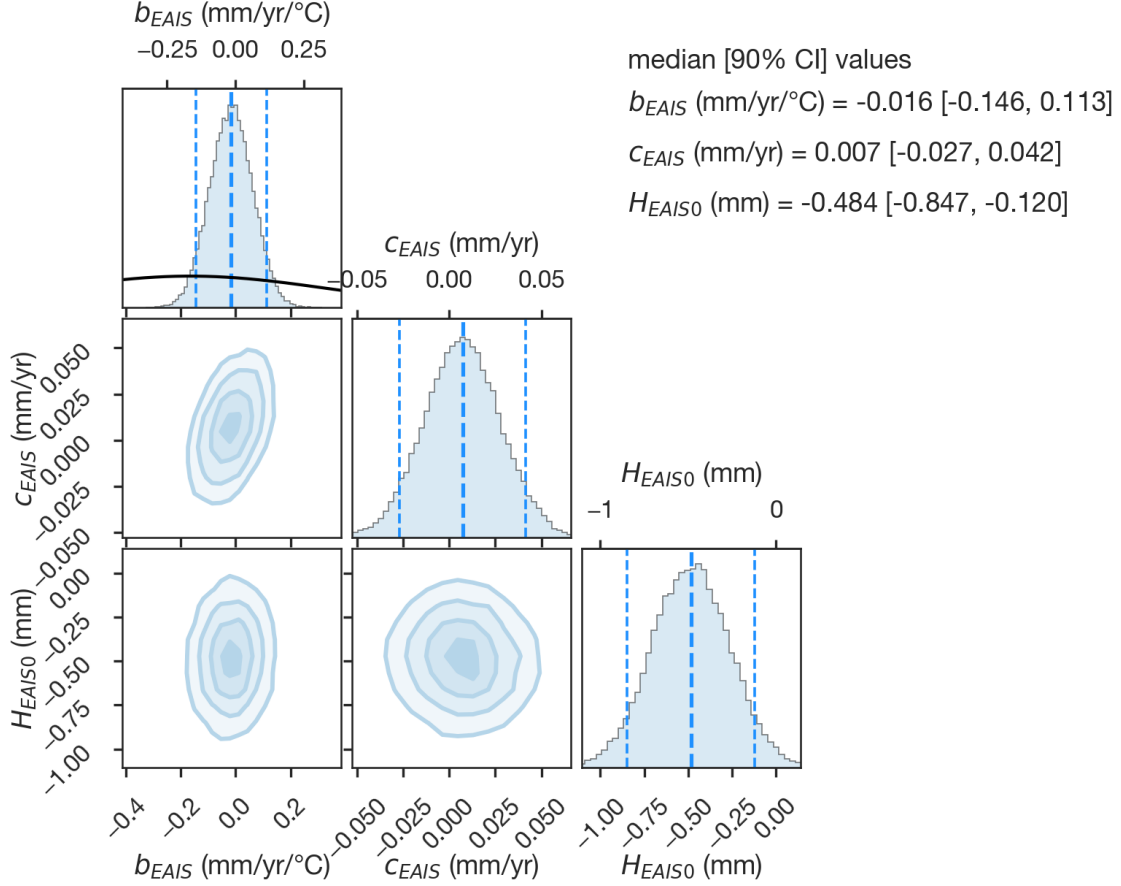


Figure S5: **EAIS model posterior correlation structure.** The joint posterior for the three parameters of the East Antarctic model fitted to the IMBIE-3 East Antarctic record (*Otosaka et al.*, 2026), with the full covariance of the cumulative record in the likelihood. This fit serves as a diagnostic, the projection instead adopting a literature surface-mass-balance sensitivity. Diagonal panels show marginal posterior histograms (pale blue), with dashed vertical lines marking the median and 90% credible interval. Off-diagonal panels show the joint posterior density, with contours denoting 1, 2, and 3 standard deviations. Only b_{EAIS} has an informative prior (black curve), centered on the literature sensitivity $C_T = 60 \pm 20 \text{ Gt yr}^{-1} \text{ °C}^{-1}$ (*Frieler et al.*, 2015; *Ligtenberg et al.*, 2013) and wide enough that the data set the posterior, while c_{EAIS} and H_{EAIS0} have wide priors that are not shown.